neutrons. If a nucleus has, for example, an even number of protons, all these protons can exist in pairs. However, if the nucleus has an odd number of protons, at least one of these protons must exist in an unpaired state. The increase in stability resulting from complete pairing in elements of even Z is responsible for their ability to accommodate a greater range of neutron numbers as illustrated for the isotopes of germanium ($_{32}$Ge, 5 stable isotopes), relative to those of gallium ($_{31}$Ga, 2 stable isotopes), and arsenic ($_{33}$As, 1 stable isotope). The same pairing stabilization holds true for neutrons so that an even-even nuclide which has all its nucleons, both neutrons and protons, paired represents a quite stable situation. In the elements in which the atomic number is even, if the neutron number is uneven, there is still some stability conferred through the proton-proton pairing. For elements of odd atomic number, unless there is stability due to an even neutron number (neutron-neutron pairing), the nuclei are radioactive with rare exceptions. We should also note that the number of stable nuclear species is approximately the same for even-odd and odd-even cases. The pairing of protons with protons and neutrons with neutrons must thus confer approximately equal degrees of stability to the nucleus. This behaviour is rather similar to the electronic structure of the atoms where also paired electrons offer a greater stability. The similarities between the nucleons in the nucleus and the electrons in the atoms is further discussed in combination with the shell model for nuclear stability, Ch. 6.

4.2. NEUTRON TO PROTON RATIO

If a graph is made (Fig. 4.1)[1] of the relation of the number of neutrons to the number of protons in the known stable nuclei, we find that in the light elements stability is achieved when the number of neutrons and protons are approximately equal ($N = Z$). However, with increasing atomic number of the element (i.e. along the Z-line), the ratio of neutrons to protons, the N/Z ratio, for nuclear stability increases from unity to about 1.52 at lead. Thus pairing of the nucleons is not a sufficient criterion for stability: a certain ratio N/Z must also exist. However, even this does not suffice for stability, because at high Z-values, a new mode of radioactive decay, α-emission, appears. Above bismuth the nuclides are all unstable to radioactive decay by α-particle emission, while some are unstable also to β-decay.

If a nucleus has an N/Z ratio too high for stability, it is said to be *neutron-rich*. It will undergo radioactive decay in such a manner that the neutron to proton ratio decreases to approach more closely the stable value. In such a case the nucleus must decrease the value of N and increase the value of Z, which can be done by conversion of a neutron to a proton. When such a conversion occurs within a nucleus, β^- (or *negatron*) *emission* is the

[1] In graphs like Fig. 4.1, Z is commonly plotted as the abscissa; we have here reversed the axes to conform with the commercially available isotope and nuclide charts.

consequence, with creation and emission of a negative β-particle designated by β^- or ${}^{0}_{-1}e$ (together with an anti-neutrino, here omitted for simplicity, see Ch. 5). For example:

$$^{116}_{49}\text{In} \rightarrow {}^{116}_{50}\text{Sn} \rightarrow {}^{0}_{-1}e^- \tag{4.1}$$

At extreme N/Z ratios beyond the so called neutron drip-line where the N/Z ratio is too high for stability, or for highly excited nuclei, *neutron emission* is an alternative to β^- decay.

If the N/Z ratio is too low for stability, then radioactive decay occurs in such a manner as to lower Z and increase N by conversion of a proton to neutron. This may be accomplished through *positron emission*, i.e. creation and emission of a positron (β^+ or ${}^{0}_{+1}e$), or by absorption by the nucleus of an orbital electron (*electron capture*, EC). Examples of these reactions are:

$$^{116}_{51}\text{Sb} \rightarrow {}^{116}_{50}\text{Sn} + {}^{0}_{+1}e^+ \quad \text{and} \quad {}^{195}_{79}\text{Au} + {}^{0}_{-1}e^- \overset{\text{EC}}{\rightarrow} {}^{195}_{78}\text{Pt} \tag{4.2}$$

Positron emission and electron capture are competing processes with the probability of the latter increasing as the atomic number increases. Beta decay is properly used to designate all three processes, β^-, β^+, and EC. (The term "beta decay" without any specification usually only refers to β^- emission.)

Thus in the early part of the Periodic Table, unstable neutron deficient nuclides decay by positron emission, but for the elements in the platinum region and beyond, decay occurs predominantly by electron capture. Both processes are seen in isotopes of the elements in the middle portion of the Periodic Table, see Figure 4.1 and Appendix C.

An alternative to positron decay (or EC) is *proton emission*, which, although rare, has been observed in about 40 nuclei very far off the stability line. These nuclei all have half-lives $\leq$1 min. For example: ^{115}Xe, $t_{½}$ (p) 18 s; proton/EC ratio, 3×10^{-3}.

We can understand why the N/Z ratio must increase with atomic number in order to have nuclear stability when we consider that the protons in the nucleus must experience a repulsive Coulomb force. The fact that stable nuclei exist means that there must be an attractive force tending to hold the neutrons and protons together. This attractive nuclear force must be sufficient in stable nuclei to overcome the disruptive Coulomb force. Conversely, in unstable nuclei there is a net imbalance between the attractive nuclear force and the disruptive Coulomb force. As the number of protons increases, the total repulsive Coulomb force must increase. Therefore, to provide sufficient attractive force for stability the number of neutrons increases more rapidly than that of the protons.

Neutrons and protons in nuclei are assumed to exist in separate *nucleon orbitals* just as electrons are in electron orbitals in atoms. If the number of neutrons is much larger than the number of protons, the neutron orbitals occupied extend to higher energies than the highest occupied proton orbital. As N/Z increases, a considerable energy difference can

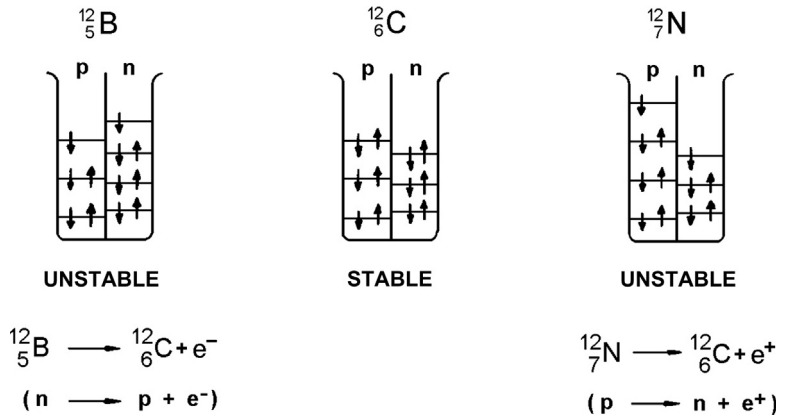


Figure 4.2 The separation and pairing of nucleons in assumed energy levels within the isobar $A = 12$. Half-life for the unstable ^{12}B is 0.02 s, and for ^{12}N 0.01 s.

develop between the last (highest energy) neutron orbital filled and the last proton orbital filled. The stability of the nucleus can be enhanced when an odd neutron in the highest neutron orbital is transformed into a proton fitting into a vacant lower energy proton orbital; see the example for $A = 12$ in Figure 4.2. In short, if the different "energy ladders" of the neutron and protons in the nucleus is investigated it becomes clear that the distance between the "steps" in the proton ladder is larger due to the repulsion force. Thus in order to achieve the lowest possible energy with a constant number of nucleons the "neuton ladder" will fill up faster thus making N/Z increase with the number of nucleons in the nucleus. These questions of nuclear forces and the energy levels of nucleons are discussed more extensively in Chapter 6.

4.3. MASS DEFECT

It was noted in Chapter 1 that the masses of nuclei (in u) are close to the mass number A. Using the mass of carbon-12 as the basis (^{1}C $\equiv$ 12 u), the hydrogen atom and the neutron do not have exactly unit masses. We would expect that the mass M_A of an atom with mass number A would be given by the number of protons (Z) times the mass of the hydrogen atom ($\mathbf{M}_\mathrm{H}$) plus the number of neutrons (N) times the mass of the neutron ($\mathbf{M}_n$), i.e.

$$M_A \approx Z\mathbf{M}_\mathrm{H} + N\mathbf{M}_\mathrm{n} \tag{4.3}$$

For deuterium with one neutron and one proton in the nucleus, we would then anticipate an atomic mass of

$$\mathbf{M}_\mathrm{H} + \mathbf{M}_\mathrm{n} = 1.007\,825 + 1.008\,665 \;=\; 2.016\,490 \text{ u} \tag{4.4}$$